

ECE 455 - Mathematical Models in Electrical Engineering

L1. Complex Variables

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Outline

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 - 1.2 Basic Operations - Polar Form
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 - 1.5 Cauchy-Riemann Equations and Harmonic Functions
 - 1.6 Exponential and Logarithmic Functions
 - 1.7 Trigonometric and Inverse Trigonometric Functions
 - 1.8 Hyperbolic and Inverse Hyperbolic Functions

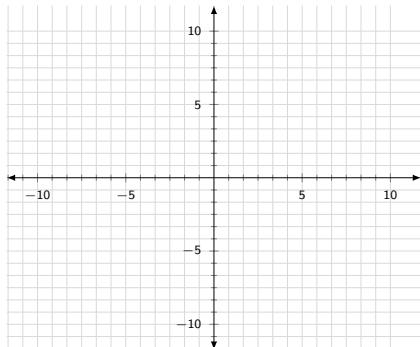
Complex Variables

Basic Operations - Rectangular Form

Complex Numbers

Given $z = x + iy$.

1. Real part: $x = \operatorname{Re}(z)$.
2. Imaginary part: $y = \operatorname{Im}(z)$.



Complex Numbers - Example

Plot the following complex numbers:

1. $z_1 = -4$

2. $z_2 = +5$

3. $z_3 = -i3$

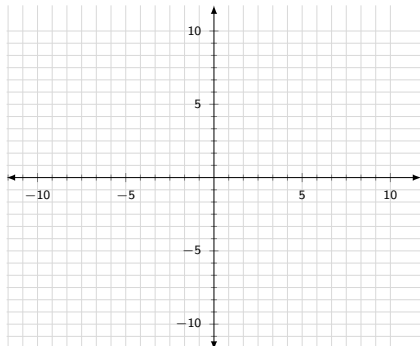
4. $z_4 = +i4$

5. $z_5 = 2 + i2$

6. $z_6 = -2 + i3$

7. $z_7 = -2 - i$

8. $z_8 = 1 - i3$

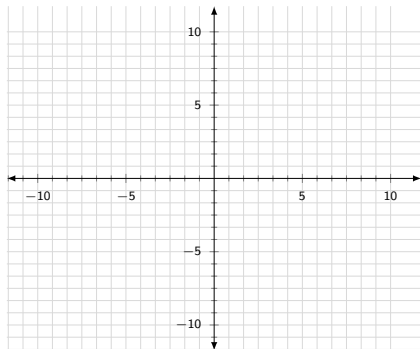


Complex Conjugate

The complex conjugate of $z = x + iy = (x, y)$ is:

$$\bar{z} = x - iy = (x, -y)$$

Geometrically, $\bar{z}$ is a reflection of z about the real axis.



Find the complex conjugate of the following complex numbers:, $z_1 = 3 - i4$, $z_2 = i2$, and $z_3 = 10$

Algebraic Operations - Rectangular Form

Take $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$$z_1 + z_2 =$$

$$z_1 - z_2 =$$

$$z_1 z_2 =$$

$$\frac{z_1}{z_2} =$$

Complex Conjugate Application

Take $z = x + iy$

$$z + \bar{z} =$$

$$z_1 - \bar{z} =$$

$$z\bar{z} =$$

Recall:

$$\frac{z_1}{z_2} =$$

Complex Conjugate Useful Properties

$$\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2$$

$$\overline{(z_1 - z_2)} = \bar{z}_1 - \bar{z}_2$$

$$\overline{(z_1 z_2)} = \bar{z}_1 \bar{z}_2$$

$$\overline{\frac{z_1}{z_2}} = \frac{\bar{z}_1}{\bar{z}_2}$$

Example

Take $z_1 = 2 + i5$ and $z_2 = -4 + i3$

Calculate $z_1 + z_2$. Show all of your work.

Example

Take $z_1 = 2 + i5$ and $z_2 = -4 + i3$

Calculate $z_1 - z_2$. Show all of your work.

Example

Take $z_1 = 2 + i5$ and $z_2 = -4 + i3$

Calculate $z_1 z_2$. Show all of your work.

Example

Take $z_1 = 2 + i5$ and $z_2 = -4 + i3$

Calculate $\frac{z_1}{z_2}$. Show all of your work.

Example

Take $z_1 = 2 + i5$ and $z_2 = -4 + i3$

Calculate $\bar{z}_1$. Show all of your work.

Example

Take $z_1 = 2 + i5$ and $z_2 = -4 + i3$

Calculate $\frac{1}{\bar{z}_2}$. Show all of your work.

Fundamental Laws

Here is assumed that z_1, z_2 , and z_3 belong to the set S of complex numbers.

Commutative Law (addition): $z_1 + z_2 = z_2 + z_1$

Associative Law (addition): $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$

Commutative Law (multiplication): $z_1 z_2 = z_2 z_1$

Associative Law (multiplication): $z_1(z_2 z_3) = (z_1 z_2) z_3$

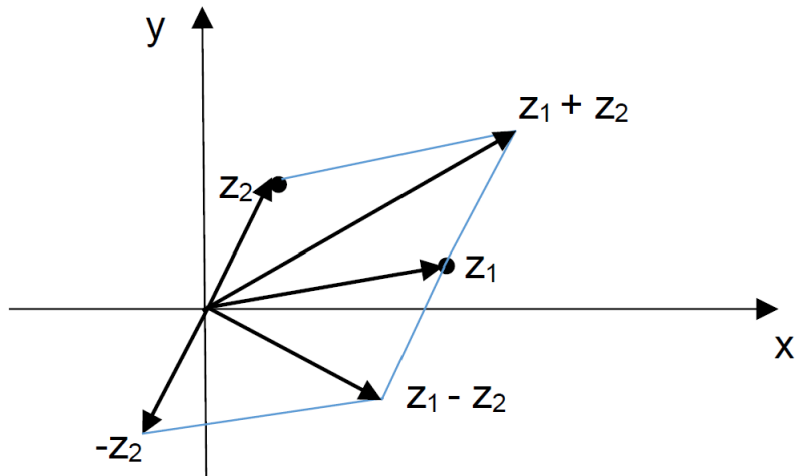
Distributive Law: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

Additive Identity: $z_1 + 0 = z_1$

Additive Inverse: $z_1 - z_1 = 0$

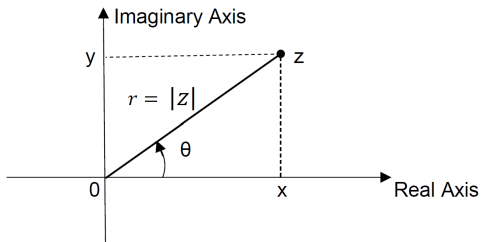
Multiplicative Inverse: $z_1 \left(\frac{1}{z_1} \right) = 1$

Graphical Representation of Complex Numbers

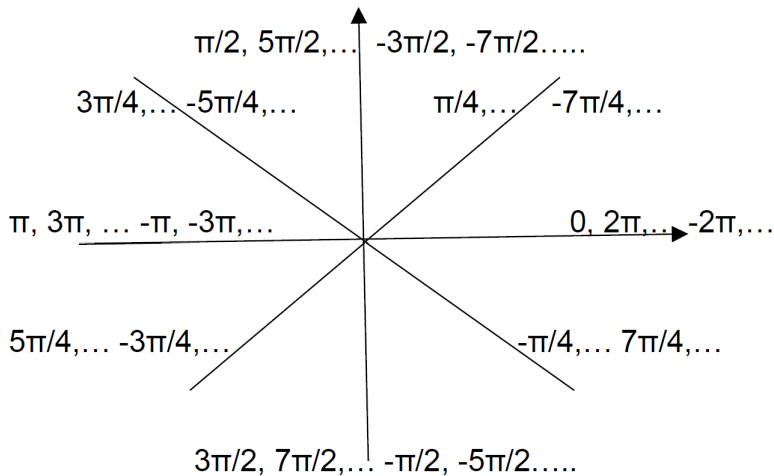


Basic Operations - Polar Form

Graphical Representation of Complex Numbers



Graphical Representation of Complex Numbers



Example

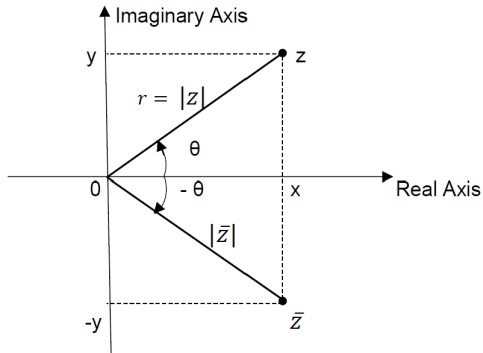
Convert the following complex number to polar form: $z = -1 + i$

Example

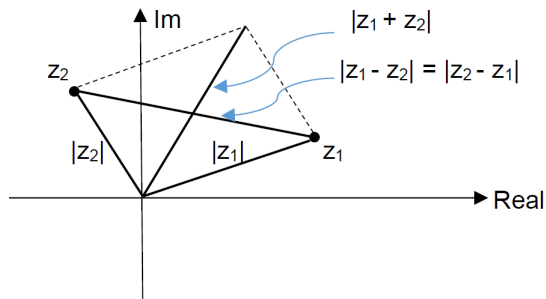
Convert the following complex number to rectangular form:

$$z = 10\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

Complex Conjugate in Polar Form



Triangle Inequality



Algebraic Operations - Polar Form

Take $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

Equality

$z_1 = z_2$ implies...

Addition

$z_1 + z_2 =$

Subtraction

$z_1 - z_2 =$

Multiplication

$z_1 z_2 =$

Algebraic Operations - Polar Form

Take $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

Integer Power ($n = 1, 2, 3, \dots$)

$$z^n =$$

Division

$$\frac{z_1}{z_2} =$$

Example

$$\text{Take } z_1 = 10\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right),$$

$$z_2 = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right),$$

$$z_3 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2},$$

$$z_4 = 2\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right),$$

$$z_5 = 1 + i$$

Evaluate $z_1 + z_4$

Example

$$\text{Take } z_1 = 10\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right),$$

$$z_2 = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right),$$

$$z_3 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2},$$

$$z_4 = 2\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right),$$

$$z_5 = 1 + i$$

Evaluate $z_1 z_2 z_3$

Example

$$\text{Take } z_1 = 10\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right),$$

$$z_2 = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right),$$

$$z_3 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2},$$

$$z_4 = 2\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right),$$

$$z_5 = 1 + i$$

Evaluate $(z_5)^{10}$

Root of Complex Numbers

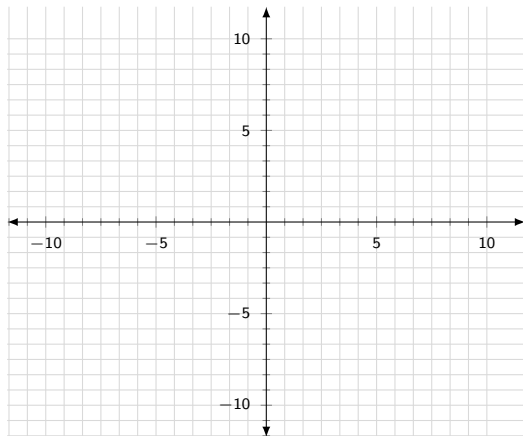
Given $w^n = z$, then $w = [z]^{\frac{1}{n}}$ is referred to as the n th root of z .

Example

Evaluate z , given that $w^4 = i16$.

Example

Plot the roots of w , given that $w^4 = i16$.



Complex Set, Function, Domain, and Range

Complex Set

Neighborhood of z_0 :

All point z that satisfy; $|z - z_0| < \varepsilon$ where $\varepsilon > 0$

Interior Points of a Set:

Points for which there exists at least a neighborhood of z_0 all whose points belong to that set.

Boundary Points of a Set:

Points for which every neighborhood of z_0 contains points that belong to that set as well as points that don't belong to that set

Exterior Points of a Set:

Points for which there exists at least a neighborhood of z_0 none of whose points belong to that set.

Connected Set:

It is a set that any two points of this set can be connected by a number of line segments all belonging to the set.

Complex Set

Simply Connected Set:

It is a connected set that every simple closed curve in the set only contains points of that set. A simple closed curve is one that does not intersect itself.

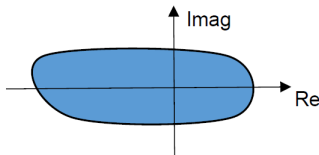


Figure 1.10 Simply Connected

Multiply Connected Set:

It is a connected set that there is at least one simple closed curve in the set with one or more points that don't belong to the set.

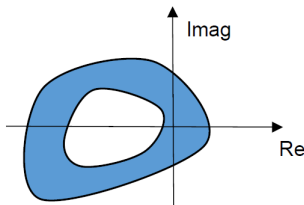


Figure 1.11 Multiply Connected

Complex Set

Bounded and Unbounded Sets:

If all the points, z in the set satisfy:

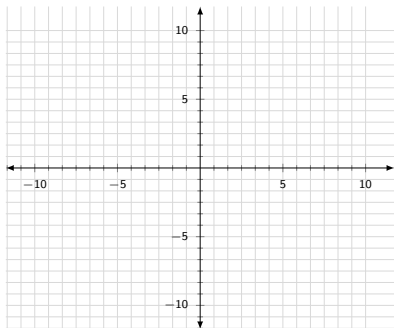
$$|z| < \rho,$$

Then the set is called bounded. If a set is not bounded, then it is called unbounded. For example, a set defined by all the points $|z| = 6$ is bounded and the set defined by $|z| > 10$ is unbounded.

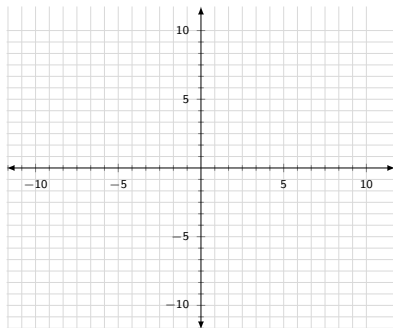
Example

Sketch the set of complex points described by the following equations:

$$|z| = 1$$



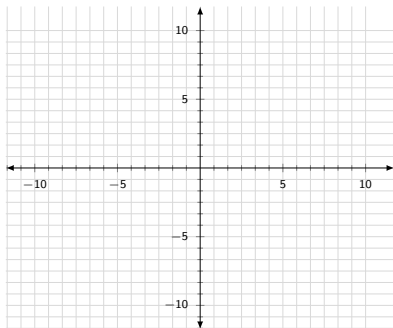
$$|z| > 1$$



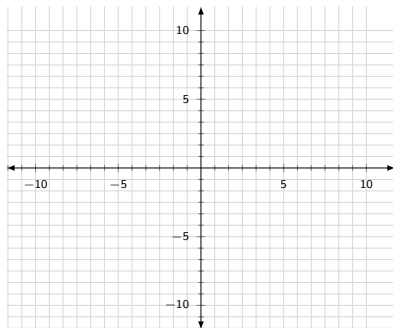
Example

Sketch the set of complex points described by the following equations:

$$|z| < 1$$



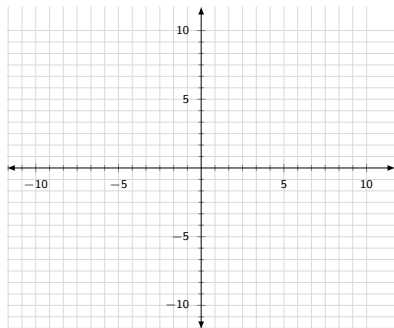
$$|z - (1 + i)| = 1$$



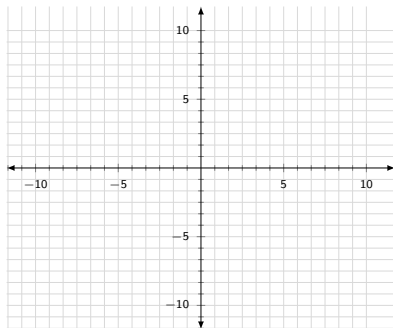
Example

Sketch the set of complex points described by the following equations:

$$2 < |z + i2| \leq 1$$



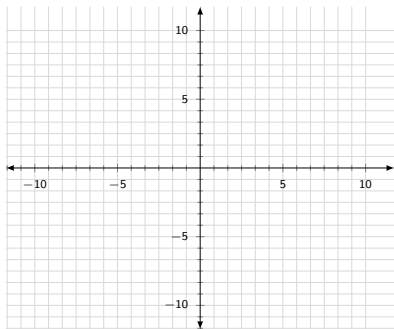
$$-1 < \text{Im}(z) < 2$$



Example

Sketch the set of complex points described by the following equations:

$$0 \leq \operatorname{Re}(z) \leq 1$$



Complex Function

A function f is a rule that assigns to each complex point, z in a set D , a complex point, w in set R .

$$w = f(z) = u(x, y) + iv(x, y)$$

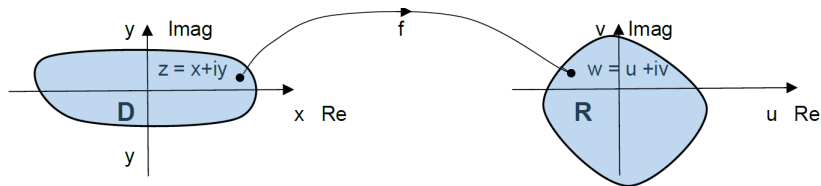


Figure 1.19 Complex Function $w = f(z) = u(x, y) + iv(x, y)$

Example

Represent the following functions in the form

$$w = f(z) = u(x, y) + iv(x, y).$$

$$w = f(z) = z^2 + 2z + 10$$

$$w = f(z) = \frac{(z+1)}{(\bar{z}-1)}$$

Example

Represent the following function,

$$w = f(z) = \frac{1}{z}, \text{ in the form}$$

$$w = f(z) = u(x, y) + iv(x, y).$$

Evaluate $f(z)$ at $z = 3 + i4$

Domain and Range

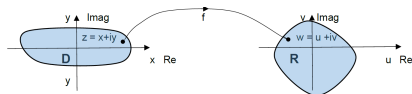


Figure 1.19 Complex Function $w = f(z) = u(x,y) + iv(x,y)$

Referring to the definition of a complex function $w = f(z)$ stated above and Figure 1.19, domain and range can be defined as:

Domain, D:

The set of allowable and permissible values of z in $w = f(z)$. D can include the entire z -plane or portion of it based on any given restriction.

Range, R:

The corresponding set of all values of w in $w = f(z)$.

Limit, Continuity, Derivative, and Analytic Function

Limit

Limit

Let $w = f(z)$ and assume this function is defined in some neighborhood of z_0 but it may or may not be defined at z_0 . We write the function $f(z)$ has a limit w_0 as z approaches z_0 .

$$\lim_{z \rightarrow z_0} f(z) = w_0 \quad (1.40)$$

and if for any real $\varepsilon > 0$, one can find a real $\delta > 0$ such that for all z satisfying $|z - z_0| < \delta$ implies $|f(z) - w_0| < \varepsilon$. Figure 1.20 shows a geometrical interpretation of this limit.

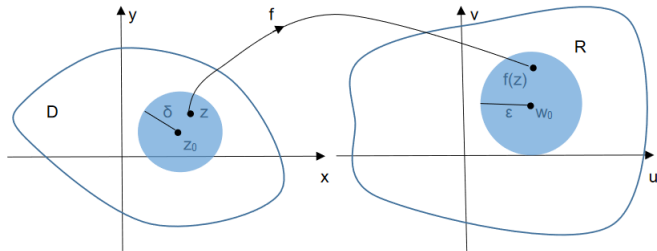
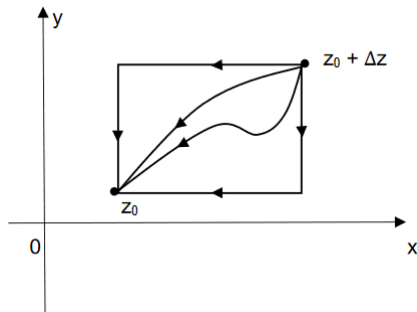


Figure 1.20 Geometrical Interpretation of Limit of a Function

Continuity and Derivative

Continuity

Derivative



Example

Differentiate,

$$w = f(z) = 5z^2 + 2z$$

Differentiate, $w = f(z) = \frac{1}{z}$

Differentiation Rules

$$F(z) = C$$

$$F'(z) = 0 \quad (1.50)$$

$$F(z) = z^n \quad n \text{ is a integer}$$

$$F'(z) = nz^{(n-1)} \quad (1.51)$$

$$F(z) = f^n(z) \quad n \text{ is a integer}$$

$$F'(z) = n f^{n-1}(z) \cdot f'(z) \quad (1.52)$$

$$F(z) = Cf(z)$$

$$F'(z) = Cf'(z) \quad (1.53)$$

$$F(z) = f_1(z) + f_2(z) + \dots + f_n(z)$$

$$F'(z) = f'_1(z) + f'_2(z) + \dots + f'_n(z) \quad (1.54)$$

$$F(z) = f_1(z) \cdot f_2(z)$$

$$F'(z) = f'_1(z)f_2(z) + f_1(z)f'_2(z) \quad (1.55)$$

$$F(z) = f_1(z)/f_2(z)$$

$$F'(z) = [f'_1(z)f_2(z) - f_1(z)f'_2(z)]/[f_2(z)]^2 \quad (1.56)$$

$$F(z) = f_1(z) \circ f_2(z) = f_1(f_2(z))$$

$$F'(z) = f'_1(f_2(z)) \cdot f'_2(z) \quad (1.57)$$

Analytic Function

A complex function $f(z)$ is analytic at a point z_0 in domain D , if $f(z)$ is defined at z_0 and is differentiable at z_0 . This definition of analyticity can be extended to an entire domain D or a subset of domain D . Clearly $f(z) = 1/z$ is analytic everywhere except at $z = 0$ and the function $f(z) = \bar{z}$ is not analytic.

Cauchy-Riemann Equations and Harmonic Functions

Cauchy-Riemann Equations in Rectangular Form

Let $w = f(z) = u(x,y) + iv(x,y)$ be differentiable at $z = x + iy$. Then partial derivatives u_x , u_y , v_x , and v_y exist and satisfy the following equations:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad (1.58)$$

Furthermore if $f(z)$ is analytic in D , then u_x , u_y , v_x , and v_y exist and satisfy equation (1.58). Equation (1.58) is called Cauchy-Riemann equations or conditions.

$$f'(z) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} = v_y - i u_y \quad (1.61)$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = u_x + i v_x \quad (1.62)$$

Example

Check if the following function $w = f(z)$ is analytic. If so, evaluate $f'(z)$ using equation (1.61) or (1.62).

Differentiate, $w = f(z) = z^3 + 2$

Differentiate, $w = f(z) = [\bar{z}]^2$

Cauchy-Riemann Equations in Polar Form

Let $w = f(z) = u(r, \theta) + iv(r, \theta)$ be differentiable at $z = r(\cos \theta + i \sin \theta)$. Then partial derivatives u_r , u_θ , v_r , and v_θ exist and satisfy the following equations:

$$r u_r = v_\theta \quad \text{and} \quad r v_r = -u_\theta \quad (1.64)$$

and

$$f'(z) = (\cos \theta - i \sin \theta)(u_r + iv_r) = (\cos \theta - i \sin \theta)(v_\theta - iu_\theta)/r \quad (1.65)$$

Furthermore if $f(z)$ is analytic in D , then u_r , u_θ , v_r , and v_θ exist and satisfy equation (1.64). Equation (1.64) is called Cauchy-Riemann equations or conditions in polar form.

Harmonic Functions

If a function $h(x, y)$ has 1st and 2nd continuous partial derivatives in the same domain D and satisfies Laplace's PDE:

$$\Delta^2 h = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = h_{xx} + h_{yy} = 0 \quad (1)$$

it is a harmonic function.

Harmonic Functions

If a function $w = f(z) = u(x, y) + iv(x, y)$ is analytic in the same domain D and the real part of w , $u(x, y)$, and the imaginary part of w , $v(x, y)$ satisfy Laplace's equation, then w is a harmonic function.

Example

Show that the real part and imaginary part of $w = f(z) = z^3 + 5z^2 + 10$ are harmonic functions.

Exponential and Logarithmic Functions

Exponential Function

Exponential Function Properties

Exponential Function Properties

Derivative of e^z

Alternate Polar Form of e^z

Example

Evaluate $f(z) = e^z$ at $z = -1 + i2$.

Example

Solve $e^z = 1 - i2$ for $z = x + iy$.

Logarithmic Function

Logarithmic Function Properties

Derivative of $f(z) = \ln(z)$

Complex Power

Example

Evaluate $(1 - i)^{1+i}$.

Trigonometric and Inverse Trigonometric Functions

Trigonometric Functions

$$e^{iZ} = \cos Z + i \sin Z \quad (1.100)$$

$$\sin Z = \frac{1}{2i} (e^{iZ} - e^{-iZ}) \quad (1.101)$$

$$\cos Z = \frac{1}{2} (e^{iZ} + e^{-iZ}) \quad (1.102)$$

$$\tan Z = \frac{\sin Z}{\cos Z} \quad (1.103)$$

$$\cot Z = \frac{\cos Z}{\sin Z} \quad (1.104)$$

$$\sec Z = \frac{1}{\cos Z} \quad (1.105)$$

$$\csc Z = \frac{1}{\sin Z} \quad (1.106)$$

$$\sin(-Z) = -\sin Z, \quad \cos(-Z) = \cos Z, \quad \tan(-Z) = -\tan Z, \quad \cot(-Z) = -\cot Z \quad (1.107)$$

Derivatives of Trigonometric Functions

$$f(z) = \sin z \qquad f'(z) = \cos z \qquad (1.108)$$

$$f(z) = \cos z \qquad f'(z) = -\sin z \qquad (1.109)$$

$$f(z) = \tan z \qquad f'(z) = \sec^2 z \qquad (1.110)$$

$$f(z) = \cot z \qquad f'(z) = -\csc^2 z \qquad (1.111)$$

$$f(z) = \sec z \qquad f'(z) = \sec z \tan z \qquad (1.112)$$

$$f(z) = \csc z \qquad f'(z) = -\csc z \cot z \qquad (1.113)$$

Derivatives of Trigonometric Function

$$\cos^2 z + \sin^2 z = 1 \quad (1.114)$$

$$\cos (z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2 \quad (1.115)$$

$$\sin (z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2 \quad (1.116)$$

$$\sin 2z = 2 \sin z \cos z \quad (1.117)$$

$$\cos 2z = \cos^2 z - \sin^2 z = 1 - 2\sin^2 z = 2\cos^2 z - 1 \quad (1.118)$$

$$\cos z = \cos x \cosh y - i \sin x \sinh y \quad (1.119)$$

$$\sin z = \sin x \cosh y + i \cos x \sinh y \quad (1.120)$$

Using equations (1.119), (1.120), and $\cosh^2 y - \sinh^2 y = 1$, we can write

$$|\cos z|^2 = \cos^2 x + \sinh^2 y \quad (1.121)$$

$$|\sin z|^2 = \sin^2 x + \sinh^2 y \quad (1.122)$$

Example

Evaluate $f(z) = \sin(z)$ at $z = -1 + i2$.

Inverse Trigonometric Functions

Considering the plus sign (principle value of $\ln$), we obtain

$$w = \cos^{-1} z = -i \ln(z + \sqrt{z^2 - 1}) \quad (1.125)$$

Similarly,

$$w = \sin^{-1} z = -i \ln(iz + \sqrt{1 - z^2}) \quad (1.126)$$

$$w = \tan^{-1} z = \frac{i}{2} \ln \frac{i+z}{i-z} \quad (1.127)$$

Derivative of Inverse Trigonometric Functions

$$f(z) = \sin^{-1}z \qquad f'(z) = \frac{1}{\sqrt{1-z^2}} \qquad (1.128)$$

$$f(z) = \cos^{-1}z \qquad f'(z) = -\frac{1}{\sqrt{1-z^2}} \qquad (1.129)$$

$$f(z) = \tan^{-1}z \qquad f'(z) = \frac{1}{1+z^2} \qquad (1.130)$$

Hyperbolic and Inverse Hyperbolic Functions

Hyperbolic Function

$$\sinh z = \frac{1}{2} (e^z - e^{-z}) \quad (1.131)$$

$$\cosh z = \frac{1}{2} (e^z + e^{-z}) \quad (1.132)$$

$$\tanh z = \frac{\sinh z}{\cosh z} \quad (1.133)$$

$$\coth z = \frac{\cosh z}{\sinh z} \quad (1.134)$$

$$\operatorname{sech} z = \frac{1}{\cosh z} \quad (1.135)$$

$$\operatorname{csch} z = \frac{1}{\sinh z} \quad (1.136)$$

$$\sinh (-z) = -\sinh z, \quad \cosh (-z) = \cosh z, \quad \tanh (-z) = -\tanh z, \quad \coth (-z) = -\coth z \quad (1.137)$$

Derivative of Hyperbolic Functions

$$f(z) = \sinh z \qquad f'(z) = \cosh z \qquad (1.138)$$

$$f(z) = \cosh z \qquad f'(z) = \sinh z \qquad (1.139)$$

$$f(z) = \tanh z \qquad f'(z) = \operatorname{sech}^2 z \qquad (1.140)$$

$$f(z) = \coth z \qquad f'(z) = -\operatorname{csch}^2 z \qquad (1.141)$$

$$f(z) = \operatorname{sech} z \qquad f'(z) = -\operatorname{sech} z \tanh z \qquad (1.142)$$

$$f(z) = \operatorname{csch} z \qquad f'(z) = -\operatorname{csch} z \coth z \qquad (1.143)$$

Properties of Hyperbolic Functions

$$\cosh^2 z - \sinh^2 z = 1 \quad (1.144)$$

$$\cosh iz = \cos z \quad (1.145)$$

$$\sinh iz = i \sin z \quad (1.146)$$

$$\cos iz = \cosh z \quad (1.147)$$

$$\sin iz = i \sinh z \quad (1.148)$$

$$\cosh (z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2 \quad (1.149)$$

$$\sinh (z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2 \quad (1.150)$$

$$\cosh (z) = \cosh x \cos y + i \sinh x \sin y \quad (1.151)$$

$$\sinh (z) = \sinh x \cos y + i \cosh x \sin y \quad (1.152)$$

Example

Evaluate $f(z) = \sinh(z)$ at $z = -2 + i2$.

Inverse Hyperbolic Functions

Considering the plus sign (principle value of $\ln$), we obtain

$$w = \cosh^{-1} z = \ln(z + \sqrt{z^2 - 1}) \quad (1.155)$$

Similarly,

$$w = \sinh^{-1} z = \ln(z + \sqrt{z^2 + 1}) \quad (1.156)$$

$$w = \tanh^{-1} z = \frac{1}{2} \ln \frac{1+z}{1-z} \quad (1.157)$$